STUDENT REPORT

DETAILS

Name

THANGADIPALLY ANVITHA

Roll Number

20R01A6753

EXPERIMENT

Title

WEEK 9B

Description

9.**Write R program to find the following Regressions with the sample data and visualize the regressions graphically.**

b) Multiple Regression

Procedure and Overvation:

20R01A6753

DESCRIPTION:

Multiple regression is an extension of linear regression into relationship between more than 2 variables.

In simple linear relation we have 1 predictor and 1 response variable but in multiple regression we have more than 1 predictor variables and 1 response variable.

The general mathematical equation of multiple regression is y=a=b1x1+b2x2+b3x3+.....+bnxn

where

y is the response variable

x1,x2,x3,.....xn are predictor variables

a,b1,b2,b3.....bn are coefficients

We create the regression model using lm() function in R. The model determines the value of the coefficients using the input data.

We can predict the value of the response variable for a given set of predictor variable using these coefficients.

lm() function

This function creates relationship model between the predictor and the response variable.

syntax:

lm(formula,data)

lm(y~x1+x2+x3+.....+xn,data)

formula is a symbol present in the relation between the response variable and predictor variable.

data is the vector on which the formula will be applied.

SOURCE CODE:

```
input<-mtcars[,c("mpg","disp","hp","wt")]

print(head(input))

#create the relationship model

model<-lm(mpg~disp+hp+wt,data=input)

#show the model

print(model)

#get the intercept and coefficients as vector values

cat("The coefficient values ","\n")

a<-coef(model)[1]

print(a)

Xdisp<-coef(model)[2]

Xhp<-coef(model)[3]

Xwt<-coef(model)[4]

print(Xdisp)

print(Xhp)

print(Xwt)
```

```r
input<-mtcars[,c("mpg","disp","hp","wt")]
print(head(input))
#create the relationship model
model<-lm(mpg~disp+hp+wt,data=input)
#show the model
print(model)
#get the intercept and coefficients as vector values
cat("The coefficient values ","\n")
a<-coef(model)[1]
print(a)
Xdisp<-coef(model)[2]
Xhp<-coef(model)[3]
Xwt<-coef(model)[4]
print(Xdisp)
print(Xhp)
print(Xwt)
```

```
> source("C:/Users/student/Desktop/20R01A6753/week9b.r")
                   mpg disp  hp    wt
Mazda RX4         21.0  160 110 2.620
Mazda RX4 Wag     21.0  160 110 2.875
Datsun 710        22.8  108  93 2.320
Hornet 4 Drive    21.4  258 110 3.215
Hornet Sportabout 18.7  360 175 3.440
Valiant           18.1  225 105 3.460

Call:
lm(formula = mpg ~ disp + hp + wt, data = input)

Coefficients:
(Intercept)         disp           hp           wt
  37.105505    -0.000937    -0.031157    -3.800891

The coefficient values
(Intercept)
   37.10551
        disp
-0.0009370091
          hp
-0.03115655
          wt
-3.800891
>
```

VIVA RESPONSES

Q)

Q)

Q)

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